VMC Vidyamandir Classes SINCE 1986

IIT JEE | MEDICAL | FOUNDATION

CODE
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JEE ADVANCED - 11 | PAPER - 1

JEE 2022

Date	Maximam Marks	Duration
27th March, 2022	180	3 Hours

General Instructions

1. The test is of **3 hours** duration and the maximum marks are **180.**
2. The question paper consists of 3 Subject (Subject I: **Physics**, Subject II: **Chemistry**, Subject III : **Mathematics**). Each Part has **three** sections (Section 1, Section 2 & Section 3).
3. **Section 1** contains **SIX (06) Multiple Correct Answers Type Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.
4. **Section 2** contains **EIGHT (08) Numerical Value Type Questions**. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08)
5. **Section 3** contains **2 Paragraphs containing TWO (02) Questions** on each paragraph**.** Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
6. For answering a question, an ANSWER SHEET (OMR SHEET) is provided separately. Please fill your **Test Code, Roll No.** and **Group** properly in the space given in the ANSWER SHEET.

Name of the Candidate (In CAPITALS) : ____________________

Roll Number : ____________________

OMR Bar Code Number : ____________________

Candidate's Signature : ____________ Invigilator's Signature ____________

MARKING SCHEME

SECTION-I

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	**+4**	If only (all) the correct option(s) is (are) chosen.
Partial Marks	:	**+3**	If all the four options are correct but ONLY three options are chosen.
Partial Marks	:	**+2**	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	:	**+1**	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	:	**0**	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	**–2**	In all other cases.

- **For Example:** If first, third and fourth are the ONLY three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

SECTION-II

- This section has **EIGHT (08)** Questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble 15the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If ONLY the correct numerical value is entered.

Zero Marks: 0 In all other cases.

SECTION-III

- This section contains **TWO (02)** paragraphs. Based on each paragraph, there are **TWO (02)** questions.
- Each question has **FOUR** options. **ONLY ONE** of these four options corresponds to the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	**+3**	If ONLY the correct option is chosen.
Zero Marks	:	**0**	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	:	**–1**	In all other cases.

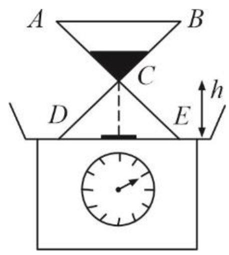
SUBJECT I : PHYSICS — 60 MARKS

SECTION 1

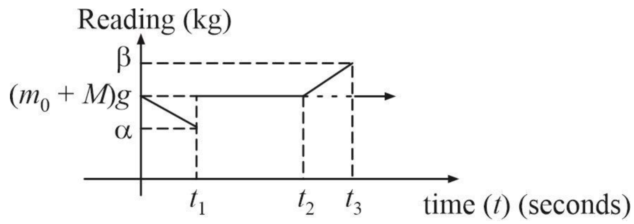
MULTIPLE CORRECT ANSWERS TYPE

This section consists of SIX (06) Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

1. An hour glass is placed on a weighing scale. Initially all the sand of mass m_0 kg in the glass is held in the upper reservoir (ABC) and the mass of the glass alone is M kg. At $t=0,$ the sand is released, It exits the upper reservoir at constant rate $\frac{dm}{dt}=\lambda$ kg/s where, m is the mass of the sand in the upper reservoir at time t sec. Assume that the speed of the falling sand is zero at the neck of the glass and after in falls through a constant height h it instantaneously comes to rest on the floor (DE) of the hour glass.



A detailed graph of reading on the scale for all times is as below. Choose the correct options.



(A) $\alpha=(m_0+M)g-\lambda\sqrt{2gh}$

(B) $\beta=(m_0+M)g+\lambda\sqrt{2gh}$

(C) $t_2=\frac{m_0}{\lambda}$

(D) $t_1=\sqrt{\frac{h}{g}}$

2. The electric potential V in volt in a region of space is given by $V=ax^2+ay^2+2az^2$ where a is a constant of proper dimensions. Work done by the field when a $2\mu C$ test charge moves from point $(0,\,0,\,0.1\,m)$ to origin is $5\times10^{-5}\,J.$ Choose the correct options :

(A) $a=1.5\times10^3$ V/m^2

(B) In every plane parallel to $x-y$ plane the equipotential lines are circles

(C) Radius of the circle of the equipotential line corresponding to $V=6250V$ and $Z=\sqrt{2}\,m$ is $1m$

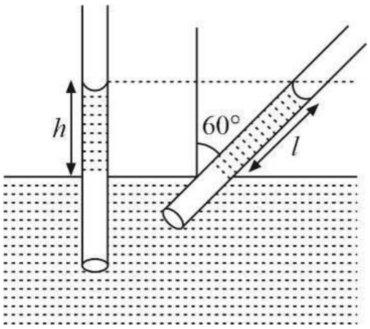
(D) $a=1.25\times10^3\,\frac{V}{m^2}$

3. A narrow glass tube of diameter 1.0 *mm* is dipped vertically in a container of water. The surface tension of water is $0.07\ N\ m^{-1}$ and the angle of contact with glass is zero. The height to which water rises in the tube (i) when tube is vertical, (ii) when tube is held slanting making an angle of 60° with the vertical is *h* and *l* respectively. Assume length of tube outside tank is 5*cm*. Two setups are used :

Setup-1 : Experiment is performed in a laboratory at rest.

Setup-2 : Experiment is performed inside an elevator accelerating up with $\frac{g}{2}$.

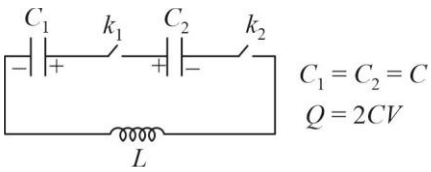
Choose the correct options. (Take : $g = 10 m/s^2$)



(A) h (setup-1) = 2.8 *cm* **(B)** l (setup-1) = 5.6 *cm*

(C) h (setup-2) = 1.87 *cm* **(D)** l (setup-2) = 3.73 *cm*

4. The capacitors C_1 and C_2 are charged to Q coulomb and V volts respectively which are connected to an inductor as shown. Keys k_1 and k_2 are closed at $t = 0$. Choose the correct options :



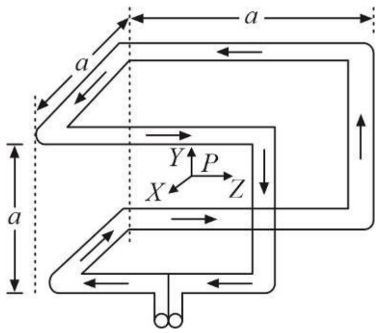
(A) Time period of oscillation is $\pi\sqrt{2LC}$

(B) Just after the key is closed EMF across the inductor is 3V

(C) When current in the circuit is maximum charge on C_1 is $\frac{3CV}{2}$

(D) Maximum charge on C_2 during the oscillation is $2CV$

5. Current-I flows around the wire frame along the edges of a cube as shown in figure. Point '*P*' is the centre of the cube. The incoming and outgoing wires have orientation towards *P*. Then, choose the correct options.



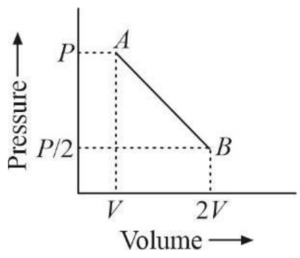
(A) The magnetic field at *P* is toward +*z* direction

(B) The magnetic field at *P* is toward –*z* direction

(C) The unit vector of magnetic field at *P* is $-\frac{1}{\sqrt{2}}(\hat{i} - j)$

(D) The magnitude of magnetic field at *P* is $B_P = \frac{2\mu_0 I}{\pi\sqrt{3}a}$

6. An ideal monoatomic gas is taken from state A (pressure P, volume V) to state B (pressure $P/2$, volume $2V$) along a straight line in the $P-V$ diagram as shown in figure. Then, choose the correct options.



(A) The work done by the gas in the process A to B exceeds the work that would be done by it if the system were taken from A to B along the isotherm

(B) In going from $A \to B$ the volume of gas where nature of process changes from endothermic to exothermic is $\frac{15}{8}V$

(C) In the $P-T$ diagram, the path AB becomes a part of a parabola

(D) In going from A to B, the temperature T of the gas first decreases to a minimum and then increases

SECTION 2

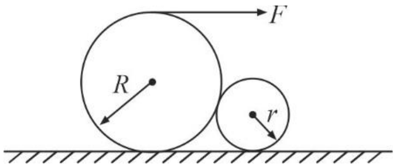
NUMERICAL VALUE TYPE QUESTIONS

This section consists of EIGHT (08) Numerical Value Type Questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08).

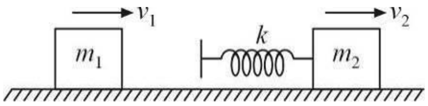
7. A block of mass 1 kg is kept at rest on a rough horizontal surface. The coefficient of friction between the block and surface is 0.5. A horizontal force of 10 N is applied on the block for 6 seconds after which the direction of force is reversed keeping the magnitude same. Find the speed (in m/s) with which the block returns to its starting point. (Take $g = 10 \text{ m/s}^2, \sqrt{3} = 1.732$)

8. A thin hollow spherical shell of radius R is cut symmetrically and its lower half is removed to get a hemispherical shell. Now the open base is covered with a circular plate of radius R made of same material and having the same thickness as the shell. The centre of mass of this covered hemisphere lies as a distance of nR from the centre of base, where n is _______ .

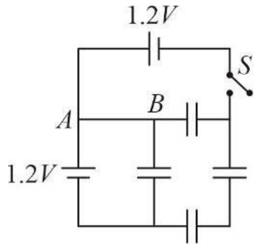
9. Two heavy cylindrical rollers of radius $R = 8\,m$ and $r = 2m$ respectively rest on a horizontal plane as shown. The larger roller has a string wound round it to which a horizontal force F can be applied as shown. Assuming that coefficient of friction μ has same value for all surfaces of contact minimum value of μ so that larger cylinder can be pulled over the smaller one is________.



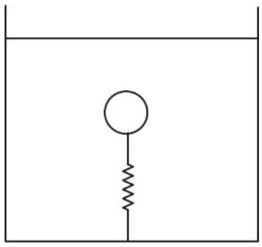
10. Two blocks of masses $m_1 = 2kg$ and $m_2 = 4kg$ are moving in the same direction with speeds $v_1 = 6\,m/s$ and $v_2 = 3\,m/s$, respectively on a frictionless surface as shown in figure. An ideal spring with spring constant k = 30000 N/m is attached to the back side of m_2. Say the moment m_1 touches the spring is $t = 0$. Kinetic energy of the system at time $t = \dfrac{\pi}{45\sqrt{10}}$ seconds is ________ Joules.



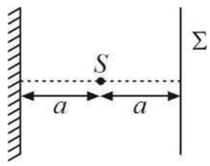
11. Find the charge (in μC) that will flow through the wire A to B if the switch S is closed. The capacitance of each capacitor shown in the figure is $C = 4.5\ \mu F$.



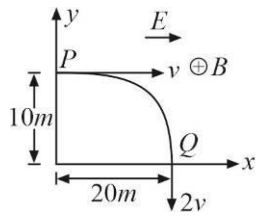
12. As a submerged body moves through a fluid, the particles of the fluid flow around the body and thus acquire kinetic energy. In the case of a sphere moving in an ideal fluid, the total kinetic energy acquired by the fluid is $\frac{1}{4}\rho V v^2$, where ρ is density of fluid, V the volume of the sphere and v is the velocity of the sphere. Consider a sphere of mass 20kg and volume $0.16m^3$ which is held submerged in a tank of water by a spring of force constant 400 N/m. Neglecting viscosity what is the period of vibration (in seconds) of the shell when it is displaced vertically and released.



13. A monochromatic point source of light S is placed in front of a perfectly reflecting mirror as shown in the figure. Wavelength of light emitted by S is λ and $a = 2.25\lambda$. Σ is screen. The ratio of intensity at the centre of screen with and without mirror is________.



14. A proton moving under the influence of a uniform electric and magnetic field, $E\hat{i}$ and $B(-k)$ follows a trajectory from P and Q as shown in figure. The velocities at P and Q are $v\hat{i}$ and $-2v\,j$. Then the value of E (in N/C) is________. ($v = 3\times10^4\,m/s$, $m_P = 6.6\times10^{-27}\,kg$ and $e = 1.6\times10^{-19}\,C$)

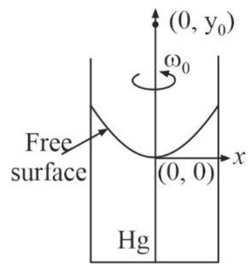


SECTION 3

This section contains 2 Paragraphs containing TWO (02) Questions on each paragraph. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

Paragraph for Q-15 to Q-16

The free surface of mercury (Hg) is a good reflecting surface. A tall cylinder partly filled with Hg is rotated about its axis with the constant angular velocity ω_0 as shown in figure. The Hg surface attains a paraboloidal profile. The radius of curvature ρ of a general profile is given by $\rho = \left|\frac{[1+(dy/dx)^2]^{3/2}}{d^2y/dx^2}\right|$, where the symbols have their usual meaning.



15. Radius of curvature ρ of Hg surface in terms of ω_0, the distance x from the cylinder axis and g is:

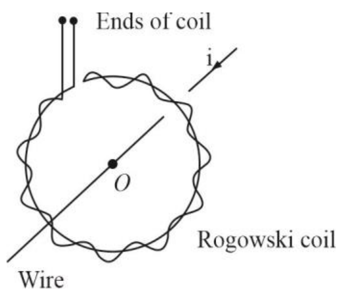
(A) $\dfrac{\left[1+\left(\dfrac{\omega_0^2 x}{g}\right)^2\right]^{3/2}}{\dfrac{\omega_0^2}{g^2}}$ **(B)** $\dfrac{(g^2+(\omega_0^2 x)^2)^{3/2}}{\omega_0^2\, g^2}$ **(C)** $\dfrac{(g^2+(\omega_0^2 x)^2)^{3/2}}{2\omega_0^2 g^2}$ **(D)** $\dfrac{(g^2+2(\omega_0^2 x)^2)^{3/2}}{\omega_0^2 g^2}$

16. Consider a point $(0, y_0)$ as shown. Condition for a real image of this object to be formed is :

(A) $y_0 < \dfrac{g}{\omega_0^2}$ **(B)** $y_0 > \dfrac{2g}{\omega_0^2}$ **(C)** $y_0 < \dfrac{g}{2\omega_0^2}$ **(D)** $y_0 > \dfrac{g}{2\omega_0^2}$

Paragraph for Q-17 to Q-18

Rogowski coil : The coil shown in the figure, called Rogowski coil, is used in electrical measurements. The coil consists of a toroidal conductor wrapped around a circular return cord. The coil can be used to measure amplitude of alternating current or value of direct current in a wire. The coil is simply clipped around the wire. Measured value of emf across the ends of the coil gives value of current amplitude, in case a.c. is flowing through the wire. To measure direct current, ends of the coil are connected to ballistic galvanometer. Current in the wire is switched off and amount of charge crossing the loop is measured by the ballistic galvanometer. This value of charge is used to calculate value of direct current in the wire.



17. Number of turns per unit length of the coil, area of each loop in the coil, resistance of the coil, are n, S and R respectively. Current flowing through the wire crossing the plane of the coil is i. A ballistic galvanometer is connected across the ends of the coil and the current is switched off, charge that flows through the coil is measured to be Q. The correct relation is :

(A) $\dfrac{\mu_0 nSi}{QR} = 1$ **(B)** $\dfrac{\mu_0 nSi}{QR} = 2$ **(C)** $\dfrac{\mu_0 nSi}{QR} = \dfrac{1}{2}$ **(D)** $\dfrac{\mu_0 nSi}{Q^2 R} = 1$

18. Consider two current carrying wires carrying currents i_1 and i_2 in opposite directions. Both the wires are perpendicular to the plane of the coil. Wire carrying currents i_1 and i_2 are respectively at a distance $0.5\,r$ and $1.5\,r$ from the centre of the coil. Reading of ballistic galvanometer depends on. [r: mean radius of the coil]

(A) i_1 **(B)** $i_1 - i_2$ **(C)** $\dfrac{i_1}{2}$ **(D)** $\dfrac{i_1 - i_2}{2}$

SUBJECT II : CHEMISTRY — 60 MARKS

SECTION 1

MULTIPLE CORRECT ANSWERS TYPE

This section consists of SIX (06) Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

1. NO_2 can be obtained by heating :

(A) NH_4NO_3 **(B)** $NaNO_3$ **(C)** $Pb(NO_3)_2$ **(D)** $LiNO_3$

2. Which of the following statement is correct with respect to metal carbonyls?

(A) As M – C π bonding increases, the C – O bond length increases

(B) As the positive charge on the central metal atom increases, the C – O bond length increases

(C) As electron density on the central metal atom increases, the C – O bond length increases

(D) The effective atomic number of the metal atom in the complex $Cr(CO)_6$ is 36

3. Based on the compounds and elements of Group 15, the correct statement(s) is/are :

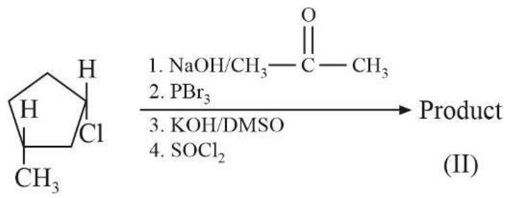
(A) d-orbitals are available for bonding in all elements

(B) H_3PO_3 disproportionate on heating

(C) The reducing nature of hydrides increases down the group

(D) P_2O_3 is more basic than Sb_2O_3

4.



(I) Reactant

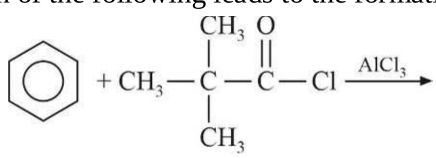
Which statement(s) are not true regarding (I) and (II).

(A) They are enantiomers **(B)** They are structural isomers

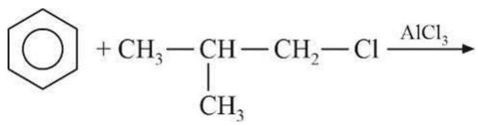
(C) They are geometrical isomers **(D)** They are diastereomers

5. Which of the following leads to the formation of tertiary butyl benzene ?

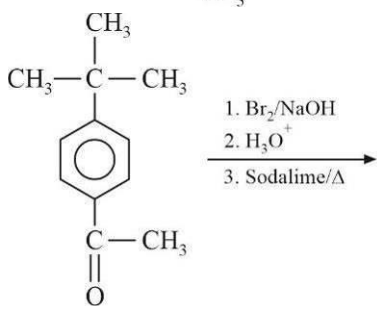
(A)



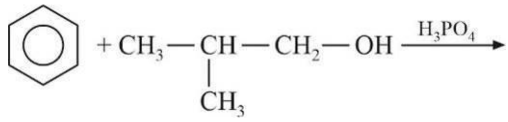
(B)



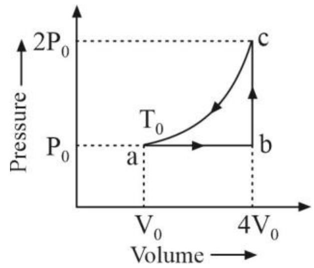
(C)



(D)



6. One mole of an ideal monoatomic gas is caused to go through the cycle shown in the figure. The correct option(s) is/are :



(A) Change in internal energy from the path a to b is $8\ RT_0$

(B) Change in internal energy from path a to c is $\frac{21}{2}RT_0$

(C) Work done by the gas from path a to b is $4\ RT_0$

(D) Change in enthalpy from path a to b is $\frac{15}{2}RT_0$

SECTION 2

NUMERICAL VALUE TYPE QUESTIONS

This section consists of EIGHT (08) Numerical Value Type Questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08).

7. Among the species given below, the total number of paramagnetic species are _______.

O_2, $O_2[AsF_6]$, KO_2, Mg, NO^+, K_2PtCl_4, $[Cu(NH_3)_4]^{2+}$, $(NH_4)_2[CoCl_4]$, $K_2Cr_2O_7$

8. If ammonia prepared by treating ammonium sulphate with calcium hydroxide is used by $CrCl_3 \cdot 6H_2O$ to form a stable co-ordination compound. Assume that both the reactions are 100% complete. If 1584 gram of ammonium sulphate and 1066 gram of $CrCl_3 \cdot 6H_2O$ are used in the preparation, the combined weight (in grams) of gypsum and the chromium-ammonia co-ordination compound thus produced is _______.

(Atomic weights in grams/mol : H = 1, N = 14, O = 16, S = 32, Cl = 35.5, Ca = 40, Cr = 52).

9. Calcium crystallises in a face-centred cubic unit cell with edge length of 0.556 nm. It contains 0.1% vacancy defects. The density of the crystal is d in gm/cm^3. The value of 100 d is _______.

(Nearest Integer) **(**Consider Avogadro's number $N_A = 6 \times 10^{23}$).

10. The E°_{cell} for the reaction $Fe + Zn^{2+} \rightleftharpoons Fe^{2+} + Zn$ is –0.33 volt at 25°C. The equilibrium concentration of Fe^{2+} is $x \times 10^{-13}$ M, when piece of iron is placed in a 1 M Zn^{2+} solution. The value of x is _______. $\left(\text{Use } \frac{2.303RT}{F} = 0.06\right)$

11. A gas expands against a variable pressure given by $P = \frac{20}{V}$ litre-atm. During expansion from volume of 1 litre to V_2 litre, the gas undergoes a change in internal energy of 400 J. Heat absorbed by the gas during expansion is 5059.8 J. The value of $15\ V_2$ is _______.

(Use $\ell nx = 2.3\log x$, 1L atm $= 101.3$J)

12. Two liquids A and B form an ideal solution. At the specified temperature, the vapour pressure of pure A is 200 mm Hg while that of B is 75 mm Hg. If the total vapour pressure above the solution is 109.13 mm Hg. The mole percent of A in the vapour phase is _______. (Nearest Integer)

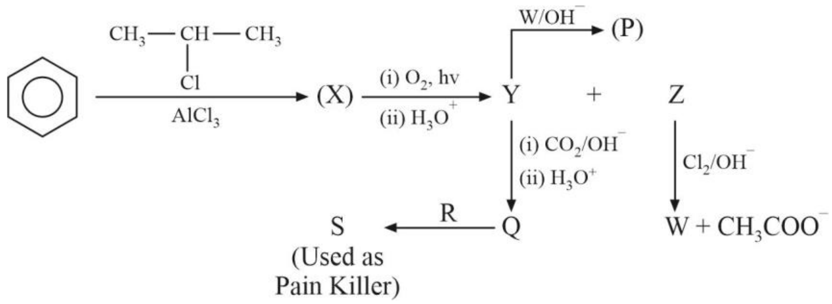
13. The solubility of MgF_2 at pH 4 is $Y \times 10^{-5}$ mol/litre. The value of Y is _______. (Given that solubility product of MgF_2 $(K_{sp}) = 6.125 \times 10^{-9}$ and the value of ionization constant of HF $(K_a) = 3.5 \times 10^{-4}$).

14. A solute S undergoes a reversible trimerisation when dissolved in a certain solvent. The boiling point elevation of its 0.1 molal solution was found to be identical to the boiling point elevation in case of a 0.08 molal solution of a solute which neither undergoes association nor dissociation. The degree of trimerisation of solute S is $Y \times 10^{-2}$. The value of Y is _______.

SECTION 3

This section contains 2 Paragraphs containing TWO (02) Questions on each paragraph. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

Paragraph for Q-15 to Q-16



Based on paragraph **'X' (Q.15, Q.16)** answer the following questions :

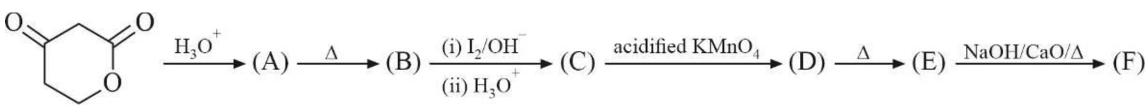
15. The compound 'R' is :

(A) $CH_3-\overset{O}{\overset{||}{C}}-Cl$ **(B)** $CH_3-\overset{O}{\overset{||}{C}}-CH_3$ **(C)** $CH_3-\overset{O}{\overset{||}{C}}-H$ **(D)** $CH_3-\overset{O}{\overset{||}{C}}-NH_2$

16. The degree of unsaturation in the compound (P) is :

(A) 4 **(B)** 3 **(C)** 5 **(D)** 6

Paragraph for Q-17 to Q-18



Based on the paragraph **"Y" (Q.17, Q.18)** answer the following questions:

17. In how many steps decarboxylation reaction is not taking place ?

(A) 2 **(B)** 5 **(C)** 4 **(D)** 3

18. The compound (F) is :

(A) CH_3-CH_3 **(B)** CH_4

(C) CH_3-CHO **(D)** $HO-\overset{O}{\overset{||}{C}}-CH_2-\overset{O}{\overset{||}{C}}-OH$

SUBJECT III : MATHEMATICS 60 MARKS

SECTION 1

MULTIPLE CORRECT ANSWERS TYPE

This section consists of SIX (06) Multiple Correct Answers Type Questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONE OR MORE THAN ONE CHOICE** is correct.

1. For a non-zero complex number z, let $\arg(z)$ denote the principal argument with $-\pi < \arg(z) \le \pi$. Then, which of the following statement(s) is(are) true ?

(A) For any two non-zero complex numbers z_1 and z_2, $\arg(z_1 z_2) - \arg(z_1) - \arg(z_2)$ is an integer multiple of 2π

(B) $\arg(\sqrt{3} - i) = -\frac{\pi}{6}$

(C) The locus of a point z such that $\arg(z) = \frac{\pi}{4}$ is : $\{z = x + ix \ \forall \ x \in R\}$

(D) $\arg\left(\frac{(1+i)(1+2i)(1+3i)}{(1-i)(2-i)(3-i)}\right) = -\frac{\pi}{2}$

2. In a triangle ABC, if the sides AB, BC and CA have lengths $5\,cm$, $5\sqrt{3}\,cm$ and $5\,cm$ respectively. Then which of the following statement(s) is (are) false?

(A) $\angle BAC = 150°$

(B) The radius of the incircle of the triangle ABC is $\frac{10\sqrt{3}+15}{2} cm$

(C) The radius of the circumcircle of the triangle ABC is 5 cm

(D) The area of the triangle ABC is $\frac{25\sqrt{3}}{4} cm^2$

3. Let $P_1 : 2x - y + z = 2$ and $P_2 : x + 2y - z = 3$ be two planes. Then, which of the following statement(s) is (are) true?

(A) Equation of the plane which passes through the point (3, 2, –1) and is perpendicular to each of the planes is $x - 3y + 5z = 2$

(B) The acute angle between P_1 and P_2 is $\cos^{-1}\frac{1}{6}$

(C) The line of intersection of P_1 and P_2 has direction ratios –1, 3, 5

(D) The equation of the plane through the line of intersection of P_1 and P_2 and the point $(1, 3, -2)$ is $17x + 4y + z = 27$

4. Let f be a real valued continuous function such that $f(x^2) = (f(x))^2$ for all x in the domain. Then which of the following is (are) correct?

(A) There is a unique function defined on [0, 1] such that $f(0) \ne 0$

(B) Let $f : (0, \infty) \to [0, \infty)$ be such that for some $a > 0$, $f(a) = 0$ then $f(x) = 0$ for all x in $(0, \infty)$

(C) There are infinitely many functions f with domain $(0, \infty)$ such that $\int_0^{\infty} f(x)\, dx < 1$

(D) There are only two such functions f with domain $(0, \infty)$ such that $\int_0^{\infty} f(x)\, dx < 1$

5. Let f be a differentiable function such that $f(f(x)) = x$ for all $x \in [0,1]$. Suppose that $f(0) = 1$, then:

(A) $f(1) = 0$
(B) f is strictly decreasing
(C) f is invertible
(D) $\int_0^1 (x - f(x))^{2020} dx = \frac{1}{2021}$

6. Let $f : [0,\infty) \to \mathbb{R}$ be a continuous function such that $f(x) = \int_0^x e^{x-t} f(t)\, dt$ for all $x \in [0,\infty)$. Then, which of the following statement(s) is (are) true?

(A) f is an exponential function
(B) $f'(x) = 2f(x)$
(C) f is a constant function
(D) f is always identically zero

SECTION 2

NUMERICAL VALUE TYPE QUESTIONS

This section consists of EIGHT (08) Numerical Value Type Questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08).

7. The value of $\left((\log_3 7)^4\right)^{\frac{2}{\log_7(\log_9 49)}} \times \left(\frac{1}{5}\right)^{\frac{8}{\log_7 5}}$ is _______.

8. The number of 5 digit numbers which are divisible by 3, with digits from the set {0, 1, 2, 3, 4, 5} given that repetition of digits is not allowed, is _______.

9. Let X be the set consisting of the first 2021 terms of the arithmetic progression 2, 6, 10, 14, …… and Y be the set consisting of the first 2021 terms of the arithmetic progression 3, 8, 13, 18, ……. Then the number of elements in the set XUY is _______.

10. If $\cos^{-1}\left(\sum_{r=1}^{\infty}\left(-\frac{x}{2}\right)^r - \sum_{r=1}^{\infty}(-x)^r\right) = \frac{\pi}{2} - \sin^{-1}\left(\sum_{r=1}^{\infty} x^{r+1} - x\sum_{r=1}^{\infty}\left(\frac{x}{2}\right)^r\right)$ then number of real solutions of the above equation in $\left(0, \frac{1}{2}\right)$ is _______

11. If $P_n = \frac{1}{n}\left((n+2)(n+4)(n+6)\ldots\ldots(n+2n)\right)^{1/n}$, where $n \in$ natural number and $\lim_{n\to\infty} P_n = L$, then $[Le]$ is ______. ($[x]$ to the greatest integer less than or equal to x).

12. $\vec{a}, \vec{b}$ and $\vec{c}$ are three vectors such that $\vec{a}\cdot\vec{b} = 0$, $|\vec{a}| = |\vec{b}| = 1$ and $|\vec{c}| = 3$. For some $m, n \in R$, let $\vec{c} = m\vec{a} + n\vec{b} + (\vec{a}\times\vec{b})$. If $\vec{c}$ is inclined at the same angle β to both $\vec{a}$ and $\vec{b}$, then the value of $9\sin^2\beta$ is _______.

13. If $\sqrt{3}p\sin x + 2q\cos x = r,\ x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ has two distinct real roots α and β with $\alpha+\beta = \frac{\pi}{3}$, where p, q, r are non-zero real numbers. Then the value of $\frac{3p}{q}$ is _______.

14. Let the area of triangle with vertices at $P(1, 1)$, $Q(0, 0)$ and $R(2, 0)$ be A_1. If area enclosed by the curve $y = x^n$, PR, QR is $0.6A_1$, then value of 'n' is :

SECTION 3

This section contains 2 Paragraphs containing TWO (02) Questions on each paragraph. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

Paragraph for Q-15 to Q-16

In the xy plane, a circle S has equation $x^2 + y^2 = 4$.

15. M_1M_2 and N_1N_2 are two chords of 'S' passing through a point $B_0(-1,-1)$ and parallel to x-axis and y-axis respectively. Let H_1H_2 be a chord of S passing through B_0 and having slope –1. If M_1M_2, N_1N_2 and H_1H_2 are chord of contact of points M_3, N_3 and H_3 respectively, then the points M_3, N_3 and H_3 lie on the curve.

(A) $x^2y^2 = 4$ **(B)** $x^2 + y^2 = 4$ **(C)** $x + y + 4 = 0$ **(D)** $(x-4)(y-4) = 4$

16. Let Q be a point on circle S. Tangent at Q to S intersect coordinate axes at points A and B, and locus of midpoint of AB is given by $\lambda_1 x^4 + \lambda_2 y^4 + x^2 + \mu y^2 = \gamma_1 x^2 y^2 + \gamma_2 x^4 y^4$.

Then $\lambda_1 + \lambda_2 + \mu - \gamma_1 - \gamma_2$ is :

(A) 5 **(B)** 3 **(C)** 2 **(D)** 0

Paragraph for Q-17 to Q-18

In a class there are four chairs C_1, C_2, C_3, C_4 and four students P_1, P_2, P_3, P_4. Initially, chair C_i is allotted to the student $P_i, i = 1,2,3,4.$ But on the examination day, these four students are randomly allotted four chairs.

17. On examination day, what will be the probability that P_2 gets previously allotted chair C_2 and NONE of the remaining students get the chair previously allotted to him/her is:

(A) $\frac{1}{12}$ **(B)** $\frac{2}{13}$ **(C)** $\frac{5}{12}$ **(D)** $\frac{11}{12}$

18. Let E_i denote event that the students P_i and P_{i-1} $(i = 2,3,4)$ do NOT sit adjacent to each other on the day of examination. Then probability of the event $E_2 \cap E_3 \cap E_4$ is :

(A) $\frac{1}{12}$ **(B)** $\frac{7}{60}$ **(C)** $\frac{2}{13}$ **(D)** $\frac{11}{12}$

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